ROTEM-guided resuscitation — Q93

Explain what each ROTEM test measures

- EXTEM: extrinsic pathway (tissue factor) – CT reflects clotting factors; MCF reflects overall clot firmness (platelets + fibrinogen)
- FIBTEM: EXTEM + cytochalasin D (blocks platelets) → fibrinogen contribution to clot
- INTEM: intrinsic pathway (heparin effect if compared with HEPTEM)
- APTEM: aprotinin/TXA added – correction compared with EXTEM indicates hyperfibrinolysis

Interpret his ROTEM results

- Low EXTEM MCF + very low FIBTEM MCF → main driver is severe hypofibrinogenaemia
- Platelet contribution = EXTEM MCF − FIBTEM MCF = 24 mm (also reduced → secondary platelet deficit)
- Normal CT → clotting factors adequate; APTEM same as EXTEM → no hyperfibrinolysis

Outline the targeted treatment

- Fibrinogen replacement first: fibrinogen concentrate 3–4 g or cryoprecipitate (target FIBTEM MCF > 10–12 mm, fibrinogen > 1.5–2 g/L)
- Then platelets if EXTEM MCF still low despite normal FIBTEM (target > 50 ×10^9/L)
- Tranexamic acid 1 g within 3 h of injury then 1 g over 8 h (CRASH-2)
- Keep Ca2+ > 1.1, temperature > 35 °C, pH > 7.2
- Repeat ROTEM after each intervention

ROTEM-guided resuscitation — Q94

Describe damage control resuscitation

- Permissive hypotension: SBP 80–90 mmHg until bleeding controlled (not in TBI: MAP ≥ 80)
- Early balanced blood products (RBC:FFP:platelets ~1:1:1, PROPPR); minimise crystalloid
- Early TXA (within 3 h)
- Rapid haemorrhage control: damage control surgery or angioembolisation (REBOA in selected)

Define the 'lethal diamond' and how to prevent each component

- Hypothermia: warm fluids, forced-air warmers, warm theatre
- Acidosis: restore perfusion, limit saline
- Coagulopathy: early plasma, fibrinogen, viscoelastic guidance
- Hypocalcaemia: monitor ionised Ca, give calcium chloride

Define massive transfusion

- ≥ 10 units RBC in 24 h, or replacement of one blood volume in 24 h
- Or ≥ 4 units in 1 h with ongoing need, or replacement of 50% blood volume in 3 h

Heparin-induced thrombocytopenia — Q95

Calculate his 4Ts score and interpret it

- Thrombocytopenia: > 50% fall, nadir ≥ 20 → 2 points
- Timing: day 5–10 → 2 points
- Thrombosis: new thrombosis → 2 points
- Other causes: none apparent → 2 points
- Total 8 (6–8 = high probability)

What should be done now

- Stop all heparin, including flushes and LMWH (cross-reactivity)
- Start a non-heparin anticoagulant: argatroban (hepatic clearance, preferred in renal failure), bivalirudin, fondaparinux or a DOAC
- Do not transfuse platelets (unless bleeding); do not start warfarin until platelets ≥ 150 (venous limb gangrene)
- Send immunoassay (anti-PF4) then functional assay (SRA/HIPA) to confirm

Explain the pathophysiology of HIT

- IgG antibodies against heparin–platelet factor 4 (PF4) complexes
- Immune complexes bind platelet Fc receptors → platelet activation
- Thrombin generation → venous and arterial thrombosis (prothrombotic despite low platelets)
- Platelets consumed and cleared → thrombocytopenia

Heparin-induced thrombocytopenia — Q96

How would you anticoagulate her for bypass

- Check antibodies: if functional assay now negative (> 100 days or negative test) heparin may be used briefly for bypass only
- If antibodies still present (acute/subacute HIT): delay surgery if possible
- If urgent: bivalirudin for bypass, or plasma exchange before heparin, or heparin + antiplatelet (cangrelor/tirofiban)
- Avoid heparin entirely after surgery (flushes, LMWH) – use non-heparin anticoagulant

Compare the non-heparin anticoagulants

- Argatroban: direct thrombin inhibitor, hepatic metabolism, monitored by aPTT, raises INR
- Bivalirudin: direct thrombin inhibitor, short half-life, partly renal
- Fondaparinux: factor Xa inhibitor, renal, long half-life
- DOACs (rivaroxaban, apixaban): used in stable patients without severe renal failure

When should warfarin be started and how long should anticoagulation continue

- Start warfarin only when platelets ≥ 150 ×10^9/L, at low dose, overlapping ≥ 5 days with a parenteral agent until INR therapeutic
- Isolated HIT (no thrombosis): anticoagulate until platelets recover (often up to 4 weeks–3 months)
- HIT with thrombosis: 3 months

Reversal of rivaroxaban — Q97

What is the reversal agent and dose

- Andexanet alfa – recombinant modified factor Xa decoy that binds factor Xa inhibitors
- Dose: high dose (800 mg bolus + 960 mg infusion) if rivaroxaban > 10 mg taken < 8 h ago; otherwise low dose (400 mg + 480 mg)
- ANNEXA-I (2024): better haemostatic efficacy than usual care in ICH, but more thrombotic events
- Alternative if andexanet unavailable: 4-factor PCC 50 U/kg (or 25–50 U/kg)

Outline the other parts of his ICH management

- BP: lower SBP to target 140 mmHg (avoid < 130) within 1 h (INTERACT-3 care bundle)
- Activated charcoal if within 2–4 h of ingestion
- Neurosurgical review; glucose, temperature control; repeat CT for expansion

How would you reverse dabigatran and warfarin

- Dabigatran: idarucizumab 5 g IV (monoclonal antibody fragment); haemodialysis removes dabigatran
- Warfarin: 4-factor PCC 25–50 U/kg (by INR) + vitamin K 5–10 mg IV
- Check drug-specific levels (anti-Xa calibrated, dilute thrombin time) when available

Reversal of rivaroxaban — Q98

How would you manage her anticoagulation before surgery

- Assess urgency vs time since last dose (apixaban half-life ~12 h; longer in renal failure)
- Check anti-Xa level calibrated for apixaban if available (< 30–50 ng/mL usually safe for surgery)
- If surgery cannot wait and level is significant: 4F-PCC 25–50 U/kg (andexanet licensed for life-threatening bleeding, not surgery in many countries)
- Tranexamic acid during surgery if bleeding

When can anticoagulation be restarted after surgery

- Prophylactic LMWH 6–12 h after surgery if haemostasis secure
- Therapeutic DOAC 24 h after low-bleeding-risk and 48–72 h after high-bleeding-risk surgery
- Balance thrombosis risk (CHA2DS2-VASc) against bleeding

Compare the pharmacology of DOACs relevant to reversal

- Dabigatran: direct thrombin inhibitor, 80% renal excretion, dialysable
- Rivaroxaban/apixaban/edoxaban: factor Xa inhibitors, highly protein-bound, not dialysable
- Standard coagulation tests are unreliable (PT may be normal on apixaban)
- Specific reversal: idarucizumab (dabigatran), andexanet (Xa inhibitors)

Haemothorax and thoracotomy — Q99

What are the indications for thoracotomy in a trauma patient with haemothorax

- Initial drainage of > 1500 mL of blood on chest drain insertion
- Ongoing drainage > 200 mL/h for 2–4 h
- Persistent haemodynamic instability or ongoing need for transfusion despite resuscitation
- This patient meets the criteria → urgent thoracotomy

What are the other life-threatening chest injuries to identify in the primary survey

- Tension pneumothorax
- Open pneumothorax
- Massive haemothorax
- Cardiac tamponade
- Flail chest with pulmonary contusion
- Airway obstruction/tracheobronchial injury

Describe the immediate resuscitation

- Large-bore IV access, activate massive transfusion protocol, balanced products
- Permissive hypotension until surgical control (in penetrating trauma)
- TXA within 3 h; avoid hypothermia
- Consider autotransfusion from the chest drain

Haemothorax and thoracotomy — Q100

State the indications for resuscitative (ED) thoracotomy

- Penetrating thoracic trauma with traumatic cardiac arrest and signs of life within the last 15 min (best outcomes)
- Penetrating non-thoracic trauma with arrest < 5 min (lower survival)
- Blunt trauma with arrest < 10 min – very poor outcome; generally not recommended
- Requires skills and equipment available (within 10 min of arrest)

What are the aims of resuscitative thoracotomy

- Release cardiac tamponade (pericardiotomy)
- Control cardiac or great vessel haemorrhage
- Cross-clamp the descending aorta to redirect blood to brain and heart
- Internal cardiac massage and defibrillation; treat air embolism

How does traumatic cardiac arrest differ from medical cardiac arrest

- Treat reversible causes first (hypovolaemia, tension pneumothorax, tamponade, hypoxia) – HOTT
- Chest compressions have lower priority than addressing causes
- Bilateral thoracostomies and rapid volume replacement with blood
- Adrenaline has a limited role in hypovolaemic arrest